

Integration — Lesson 30 Test: Integration by Parts I

10 questions · show your work in the box · no calculator · leave answers exact (e , π , $\ln$) · always check by differentiating

Name: _____ Date: _____ Score: _____ / 10

1. Set up the parts only — do NOT compute the integrals. For each, state u , dv , du , and v , ^{1 pt} and name the LIATE rung that u came from.

(a) $\int x e^{4x} dx$ (b) $\int x \sin(3x) dx$

2. Compute $\int (x - 3)e^x dx$ using the full ritual (state u , dv , du , v ; assemble; simplify). ^{1 pt}
Then check your answer by differentiating.

3. Compute $\int (x - 1)\cos x dx$, and check by differentiating. (Write every minus sign out ^{1 pt} — do not do the signs in your head.)

4. Compute $\int 4x \sin x \, dx$. Pull the constant out front first, then use by parts. Check by differentiating. 1 pt

5. Explain why u-substitution cannot finish $\int x e^{4x} \, dx$. Show what happens if you try $u = 4x$, then say which differentiation rule this integrand came from and why that tells you to use by parts. 1 pt

6. Now finish the integral you set up in question 1(a): compute $\int x e^{4x} \, dx$. Check by differentiating. (Watch where each factor of $1/4$ comes from.) 1 pt

7. Compute $\int x \sin(3x) \, dx$, and check by differentiating. (Two separate sources of minus signs — track them one at a time.) 1 pt

8. A student attacks $\int x \cos x \, dx$ with the choice $u = \cos x$, $dv = x \, dx$. Carry that choice through one full step. Then: (a) is the equation the student wrote actually false, or just unhelpful? (b) What is the tell-tale sign the choice was the wrong direction? (c) State the choice that works and give the correct answer. 1 pt

9. Evaluate $\int_0^1 x e^{2x} \, dx$ exactly (leave e in your answer). Then say in one sentence why the answer must come out positive. 1 pt

10. Challenge. (a) Compute $\int (2x + 1)e^{-x} dx$ and simplify it into the form (something) $\cdot e^{-x} + C$. (b) Check part (a) by differentiating. (c) Use your antiderivative to evaluate $\int_0^1 (2x + 1)e^{-x} dx$ exactly.

1 pt
